

VisionBeam: Vision-Driven Camera Interface for Autonomous Stage Lighting

Ryan Quinn
MIT

rquinn1@mit.edu

Tye Phoenix
MIT

tphoenix@mit.edu

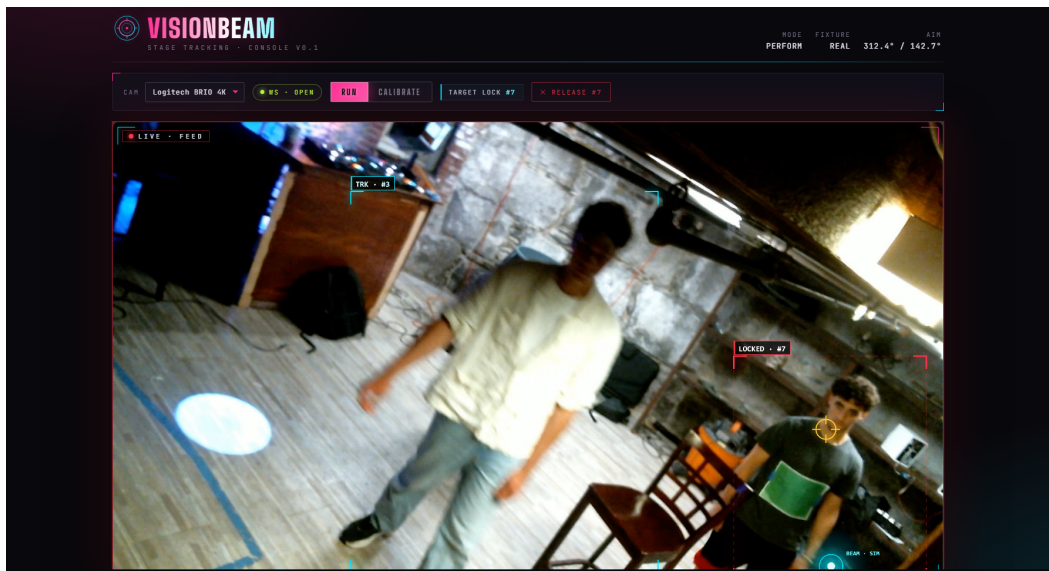


Figure 1. The VisionBeam operator console in run mode. The detector returns two persistent tracks (cyan corner brackets) and the operator has locked onto the right-hand subject (red brackets), causing the live target reticle and the simulated DMX beam to follow that ID. The header surfaces the active mode, fixture connection, and current pan/tilt aim; the controls strip exposes camera selection, the run/calibrate toggle, and a one-click release for the locked target.

1. Abstract

This paper introduces VisionBeam, an autonomous stage lighting framework that identifies and tracks the most active performer through a hybrid approach combining deep learning object detection with classical motion analysis. Traditional optical flow methods fail under extreme illumination changes, while standard object detectors lack the temporal awareness needed to assess activity levels. Designed for low-latency deployment on standard hardware, VisionBeam runs on a laptop CPU, outperforms brightness-constancy baselines (frame differencing and Farneback flow) on targeting accuracy, and degrades less than detection-only tracking under dynamic lighting while running roughly $1.8\times$ faster. We evaluate across four controlled lighting conditions, from ambient room light to a moving stage fixture combined with dynamic exter-

nal wash lights, against three baselines: raw frame differencing, Farneback dense flow, and detection-only tracking. Hybrid’s accuracy degrades by 48% from ambient to the worst condition, which is the smallest relative degradation of the four methods. Masking the motion heatmap with detected bounding boxes is the mechanism behind this stability: under the hardest lighting condition it cuts jitter to less than half of unmasked frame differencing. The complete source code and evaluation datasets are provided at github.com/rquinnmit/visionbeam.

2. Introduction

The automation of stage lighting has traditionally relied on either pre-programmed cues executed by a lighting operator or hardware-based tracking systems where performers wear body-mounted tracking hardware (e.g. infrared emitters or ultra-wideband tags). While effective, these systems are ex-

pensive, require significant setup, and are impractical for unstructured environments like dance clubs or interactive art installations where participants cannot reasonably wear tracking hardware. Vision-based autonomous lighting offers a compelling alternative, promising to lower the barrier to entry for dynamic live events by using standard cameras to steer lighting fixtures.

The core challenge in vision-based stage lighting is target selection: determining *where* to point the light in real time. In a dance environment, the system should ideally follow the most active participant. The stage is a hard place to do that. Dynamic lighting produces a lot of non-human motion: fog catches a moving beam and turns into a slow volumetric blob, strobes flip the whole image brightness in a single frame, and a sweeping beam leaves a bright high-contrast streak across the floor that any motion estimator will happily lock onto.

Classical computer vision techniques for motion estimation, such as optical flow and frame differencing, operate on the assumption that a given patch of the image stays about equally bright from frame to frame. Under stage lighting, this assumption is violated. These methods frequently fail by locking onto the moving beam of light itself or the flashes of a strobe rather than the human performers. On the other hand, modern deep learning object detectors (such as YOLO) are remarkably robust at identifying humans even under poor lighting conditions. However, bounding boxes alone lack temporal motion context; a detector cannot easily differentiate between a person standing perfectly still and a person dancing energetically without the addition of complex, high-latency pose estimation or action recognition networks, which struggle to run in real-time on venue hardware.

To address these challenges, we introduce VisionBeam, a hybrid tracking framework built for dynamic stage environments. VisionBeam uses a deep learning person detector to produce per-frame bounding boxes, then restricts all motion analysis to the pixels inside those boxes. This reduces false motion from strobes, color sweeps, and beams that would otherwise dominate full-frame optical flow or frame differencing. Within each bounding box, VisionBeam measures motion by differencing the current and previous downsampled grayscale frames, blurs the result into a heatmap, and selects the location of the heatmap’s peak as the aim target. A fitted quadratic regression then maps that pixel coordinate directly to fixture pan/tilt angles with no intermediate smoothing. Restricting the motion signal to detected people gives us most of what an action recognizer would, without the runtime cost. The whole pipeline still runs in real time on a laptop CPU.

To validate VisionBeam, we compare it against three baselines: raw frame differencing, Farneback dense optical flow, and detection-only tracking. We test all four methods

across a spectrum of lighting conditions. We also measure when the classical baselines start to fail and how much accuracy and stability the hybrid design recovers. This work contributes a practical approach to autonomous target tracking in the chaotic visual environments of live entertainment, prioritizing robustness to dynamic lighting and real-time performance over absolute pixel-level accuracy.

3. Related Work

We draw on three lines of work: classical motion estimation, deep person detection, and autonomous stage lighting.

3.1. Optical Flow and Illumination Robustness

Classical optical flow methods, such as the dense flow algorithm proposed by Farneback [1], rely heavily on the brightness constancy assumption. Surveys by Fortun et al. [2] and Alfarano et al. [3] point to rapid illumination changes as the dominant failure modes for these classical estimators. Kim et al. [4] proposed frameworks to handle both multiplicative and additive illumination changes, demonstrating that brightness constancy fails under large illumination variations. Mileva et al. [5] take a different angle: instead of matching raw pixel brightness, they match photometric invariants that stay stable under global lighting shifts. These methods help, but the extreme, localized lighting artifacts common in stage environments still slip through. Deep learning approaches, such as RAFT [6], achieve state-of-the-art optical flow performance but are generally too computationally expensive for real-time venue deployment without specialized hardware. Agnihotri et al.’s FlowBench [7] adds another caveat: even big-dataset models lose robustness on out-of-distribution inputs, and stage lighting is firmly that.

3.2. Deep Learning in Low-Light Environments

Modern object detectors are surprisingly resilient to bad lighting. The YOLOv8 architecture [8] is a strong real time baseline, and research into low-light detection has produced more specialized architectures such as Dark-YOLO [9] and POP-YOLOv8 [10] that improve performance under dark or nighttime conditions. These works show that deep learning models can reliably identify people even in poor lighting. Loh and Chan [11], in their paper introducing ExDark (a dataset comprised of low-light images), note that low light affects feature representations at a deep level, so illumination invariance is rarely enough on its own. Rather than training a specialized low-light detector, we use a standard YOLOv8 model and rely on its built-in structure to filter out non-human artifacts before they reach the motion stage.

3.3. Multi-Object Tracking

Identifying performers persistently is very important for autonomous lighting. It allows the system to smoothly fol-

low an individual or hand off between targets. We utilize ByteTrack [13], a multi-object tracker that links detections across frames so that identities persist through occlusions and tracking failures. This is extremely relevant in dance environments, where constant motion and uniform appearances make association particularly difficult [12]. Thus, the combination of YOLOv8 and ByteTrack provides a stable foundation for identifying and tracking performers.

3.4. Autonomous Stage Lighting

Commercial systems like Naostage K SYSTEM use multi-camera 3D tracking with proprietary models, but they need specialized hardware and keep the tracking infrastructure separate from the lighting console. Gao et al. [14] are closest to us in scope: a CNN paired with a particle filter that identifies and positions actors. Their setting does not include the fixture’s own beam confusing the camera, which is a large part of what makes ours hard.

Across these threads, no existing approach explicitly separates human motion from lighting-induced motion with a cheap, real-time signal: commercial systems offload the work to specialized multi-camera hardware, classical flow-based methods assume the brightness constancy that stage lighting violates by design, and detection-only methods discard motion entirely. VisionBeam targets this gap by using person detection as a semantic mask over a frame-differencing heatmap, which keeps the motion signal cheap while suppressing the artifacts that defeat the classical approaches.

4. Methodology

We built an end-to-end autonomous lighting pipeline that turns camera frames into fixture movements in real time. The pipeline has four stages: spatial calibration, target selection, spatial translation, and hardware actuation. Our main contribution is the target selection stage, where we introduce the VisionBeam hybrid pipeline and compare it against three baseline methods. The investigation centers on whether masking a cheap frame-differencing signal with detected person boxes preserves classical-motion latency while suppressing the lighting artifacts that defeat brightness-constancy methods.

4.1. Formulation and Notation

Let $V = I_1, I_2, \dots, I_T$ be the camera stream, with each frame $I_t \in \mathbb{R}^{H \times W \times 3}$ an RGB image at time t . The system needs to emit a corresponding stream of pan and tilt angles, $(\theta_{pan}^t, \theta_{tilt}^t)$, aimed at whoever is most active in the frame.

We split this problem into two steps. A target selection function f maps the recent frame history to a 2D pixel coordinate:

$$f(I_t, I_{t-1}, \dots) \rightarrow (u_t, v_t)$$

A spatial translation function g then maps that pixel coordinate to physical fixture angles:

$$g(u_t, v_t) \rightarrow (\theta_{pan}^{(t)}, \theta_{tilt}^{(t)})$$

In our deployment, g is a pair of learned quadratic regressions $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, one per axis. The fixture sits at a fixed, calibrated position above the performance area.

4.2. System Architecture

The three supporting stages around the target selection module are spatial calibration, spatial translation, and hardware actuation.

Spatial Calibration: The pixel-to-angle mapping is built once per venue through a browser-based calibration UI, shown in Figure 1. The operator drives the fixture to a sequence of pan/tilt setpoints, each chosen so that the projected beam lands at a different location on the floor, and clicks in the camera feed where the beam dot actually appears. Each click records a $(p_x, p_y, \theta_{pan}, \theta_{tilt})$ correspondence. After at least six in-frame samples have been collected, the system fits two quadratic least-squares regressions (one per axis) of the form

$$\theta = c_0 + c_1 p_x + c_2 p_y + c_3 p_x p_y + c_4 p_x^2 + c_5 p_y^2$$

via `numpy.linalg.lstsq` and persists the resulting coefficient vectors to disk. The quadratic basis is intentional, since a fixture mounted off-axis from the camera introduces perspective and radial distortion that a purely linear fit cannot absorb.

Spatial Translation: At runtime, each target pixel (u_t, v_t) produced by the tracker is converted to fixture angles by evaluating the two saved regressions:

$$\theta_{pan}^{(t)} = a^\top \phi(u_t, v_t)$$

$$\theta_{tilt}^{(t)} = b^\top \phi(u_t, v_t)$$

where $\phi(p_x, p_y) = (1, p_x, p_y, p_x p_y, p_x^2, p_y^2)$ is the same quadratic feature vector used during calibration and $a, b \in \mathbb{R}^6$ are the fitted coefficient vectors. There is no floor projection, no inverse kinematics, and no temporal smoothing on the live path, so the entire spatial transform from tracker output to commanded angle reduces to a single dot product per axis.

Hardware Actuation: The pan and tilt angles are scaled to 16-bit values, each split into a coarse and fine 8-bit DMX channel, and sent to the fixture over a USB-to-DMX512 serial interface running at 250 kBaud with 8-N-2 framing. A background thread continuously transmits the current DMX universe at approximately 40 Hz, so the fixture always has the latest commanded angles.

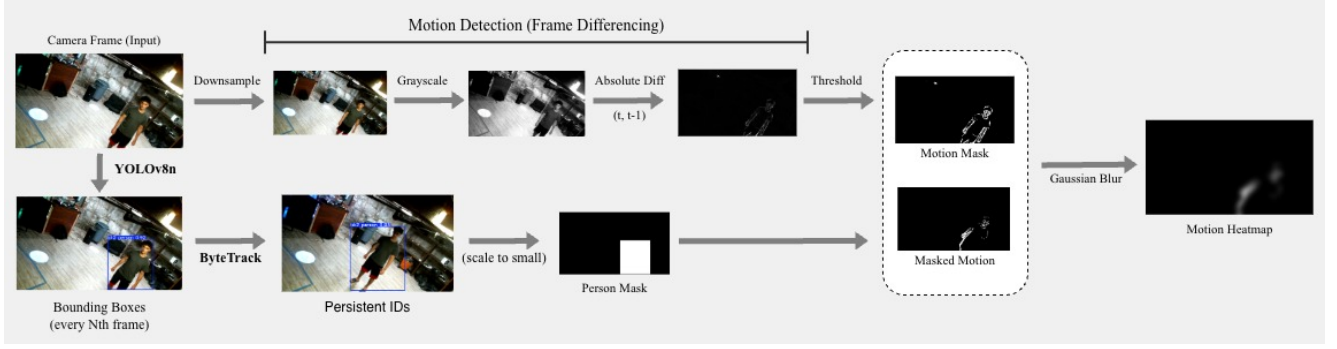


Figure 2. Hybrid tracking pipeline. Each incoming camera frame is processed along two parallel paths. The upper path performs motion detection via temporal frame differencing. The frame is downsampled to 320 pixels wide, converted to grayscale, differenced against the previous grayscale frame, and thresholded to produce a binary motion mask. The lower path runs YOLOv8n person detection (every Nth frame) followed by ByteTrack identity association, producing bounding boxes with persistent IDs. These boxes are scaled down and rasterized into a person mask. The two masks are combined to suppress motion outside of tracked people. The result is Gaussian-blurred into a smooth motion heatmap whose peak defines the target pixel.

4.3. Hybrid Tracking

The hybrid tracking module (outlined in Figure 2) pairs a deep learning person detector with classical frame differencing so that the motion signal only ever comes from regions that actually contain people.

Person Detection and Tracking: YOLOv8n detects people (class 0, confidence threshold 0.3), and ByteTrack assigns persistent IDs across frames so that bounding boxes survive brief occlusions. The Ultralytics implementation provides ByteTrack natively. To save compute, the detector runs every Nth frame (default $N = 2$) and the tracker’s boxes are reused in between. The output is a set of tracked bounding boxes $B = b_1, b_2, \dots, b_k$.

Frame Differencing: In parallel, the frame is downsampled to 320 pixels wide and converted to grayscale. The system computes the absolute difference between the current downsampled frame I_t and the previous downsampled frame I_{t-1} , then thresholds the result into a binary motion mask M_{raw} . Working at a lower resolution keeps the per-frame cost low and tolerates small registration errors.

Person Masking: A binary mask M_{person} is built from the bounding boxes (rescaled to match the downsampled frame). Pixels inside any box are set to 1, all others to 0. The motion mask is filtered with a bitwise AND, $M_{filtered} = M_{raw} \odot M_{person}$, which zeros out any motion outside of a person (e.g. a strobe reflection on a wall or moving fog).

Beam Masking: To stop the fixture from following its own light, the pipeline records the previous frame’s aim point and zeros out a circular region of radius r around it in a beam mask M_{beam} . The final motion mask is $M_{final} = M_{filtered} \odot M_{beam}$.

Target Selection: A minimum area gate suppresses invalid output when the masked motion is too small. If fewer

than A_{min} pixels are active in M_{final} , the method returns no new target and the fixture holds its previous aim. Otherwise, M_{final} is blurred with a Gaussian filter to produce a smooth heatmap, and the target (u, v) is the location of the heatmap’s peak, rescaled back to original frame coordinates.

The result is a target signal that responds only to human motion. The detector filters out lighting artifacts that would otherwise dominate a motion heatmap. The motion signal inside each box gives a per-region activity measurement without requiring pose estimation or action recognition.

4.4. Baseline Comparisons

To validate VisionBeam, we compare it against three baseline methods. Each method takes a sequence of frames and outputs a 2D pixel coordinate for the desired aim point.

Frame Differencing (Classical Motion): This baseline uses only classical motion detection with no understanding of what is in the scene. It takes the absolute difference between consecutive grayscale images, thresholds the result into a binary motion mask, and blurs that mask with a Gaussian filter to produce a heatmap. The target is the peak of the heatmap. The method is cheap to run, but it reacts to any source of motion.

Farneback Dense Flow (Classical Flow): This method uses Farneback’s algorithm to estimate dense optical flow between consecutive frames. We compute the magnitude of the flow vectors at each pixel, blur the magnitude map, and pick its peak as the target. Dense flow handles global brightness changes better than simple frame differencing, but it is still fooled by the localized, high-contrast moving artifacts typical of stage lighting.

Detection Only (Deep Learning): This baseline relies only on deep learning. YOLOv8n detects people, and Byte-

Track assigns each one a persistent ID. The target is the center of the bounding box belonging to the track with the lowest ID. This corresponds to the person who has been visible the longest, as ByteTrack assigns IDs in order of appearance. If no track IDs are available yet, the system falls back to the first detection in the model’s output. Lighting artifacts do not affect this method, but it has no motion signal at all, so it cannot tell the difference between a stationary person and someone who is actively moving.

4.5. Performance Metrics

We evaluate the performance of each method using the following metrics:

Targeting Accuracy (Mean Error): The primary metric is the mean Euclidean distance in pixels between the predicted target location and the ground truth marker location.

Target Stability (Jitter): We measure jitter as the total path length traveled by the predicted target per second (pixels/second). High jitter indicates an erratic, unusable control signal.

Robustness Drop-off: We calculate the percentage increase in targeting error from the baseline ambient lighting condition to the most challenging conditions. This highlights how severely a method degrades under stress.

Processing Speed: We measure the average frames per second (FPS) achieved by each method on the evaluation hardware.

5. Experimental Results

To evaluate the target selection methods, we constructed a controlled dataset and an offline evaluation framework. The framework runs every method on every clip and reports the metrics listed in section 4.5.

5.1. Dataset and Ground Truth

We recorded twenty 30-second video clips (four lighting conditions x five repetitions per condition). We used a fixed Logitech webcam, attached to the base of the lighting fixture, with 1920x1080 resolution and 30 FPS. We used a 15 channel, 120W ring spot lighting fixture with pan and tilt ranges of 540° and 270°, respectively. Each clip features one subject wearing a bright green marker as the ground-truth reference, and the marker moves with the subject through a representative range of motion. Another subject and chairs (shown in Fig 3) were introduced to test occlusions in the space where the initial subject was moving.

The four conditions were as follows:

1. **Ambient:** Standard room lighting with no external washers or fixtures.
2. **External (Static):** A set of external static lights alongside the standard room lighting.

3. **External (Dynamic):** A dynamic external light alongside the standard room lighting.

4. **Fixture + External (Dynamic):** Dynamic external lighting, standard room lighting, and dynamic lighting from the VisionBeam light fixture.

Per-frame ground truth is extracted using HSV thresholding tuned for the green marker. Frames where the marker is occluded or fails detection are linearly interpolated from the nearest detected neighbors. A typical clip retains a detection rate above 95%.

The marker’s pixel position is the canonical ground truth. All predicted-versus-marker comparisons are performed directly in pixel space, so no floor projection is involved. Because the marker is worn on the chest rather than at the feet, the absolute error magnitudes contain a geometric offset against any method whose target lands away from the torso (see Section 6).

5.2. Evaluation Configuration

Before recording the dataset, we calibrated the fixture to the studio using the procedure in Section 4.2; the fitted mapping is what allows the VisionBeam light to actually aim when the fixture is active, but it is not used by the evaluation itself, which compares predictions to the chest marker directly in pixel space.

Our evaluation framework calls each method’s `process_frame` function frame-by-frame on the raw video and records the predicted target. The deployed pipeline applies no temporal smoothing between the tracker’s pixel output and the quadratic pixel-to-angle mapping that drives DMX, so the evaluation’s accuracy and jitter numbers directly characterize the signal that controls the fixture in the field.

5.3. Targeting Accuracy

One caveat before Table 1: the chest-worn marker sits near a bounding-box center but away from the body parts that Hybrid’s heatmap peak tends to track, so absolute error favors detection-only by construction. The robustness and speed metrics later in this section are less sensitive to this offset.

Detection-only achieves the lowest absolute targeting error in every condition (Table 1), consistently 85-115 pixels better than Hybrid. The gap holds on 19 of 20 individual clips. Our hypothesis is that the reasoning is structural rather than a failure of hybrid under any specific lighting. Detection-only returns the geometric center of a tracked bounding box (which sits near the green marker in pixel space), while Hybrid returns the peak of a motion heatmap that lands on whichever body part is moving fastest. Additionally, Hybrid noticeably outperforms the classical frame differencing and Farneback optical flow methods when it comes to targeting accuracy. Figure 5 displays how accurately each method tracks the target in an ambient scene.

Condition	Frame Differencing	Farneback	Detection-only	Hybrid (ours)
Ambient	352.9 $\pm$ 114.0	363.9 $\pm$ 128.1	227.4 $\pm$ 106.7	314.9 $\pm$ 105.0
External (Static)	328.2 $\pm$ 83.3	340.2 $\pm$ 92.2	214.8 $\pm$ 124.2	300.8 $\pm$ 86.9
External (Dynamic)	424.6 $\pm$ 96.1	432.9 $\pm$ 97.5	285.1 $\pm$ 102.1	385.2 $\pm$ 89.9
Fixture + External (Dynamic)	601.1 $\pm$ 91.6	590.1 $\pm$ 65.3	351.5 $\pm$ 82.3	465.2 $\pm$ 95.9

Table 1. Mean targeting error in pixels (where lower is better). Detection-only exhibits the lowest targeting error among all four conditions, while Hybrid exhibits the second lowest targeting error among the same four conditions. Bold marks the best method in each row. We took the mean $\pm$ the sample standard deviation.



Figure 3. Obstacles were introduced in several recording sets to test how our hybrid and baseline tracking methods respond to occlusions of the desired subject, who is wearing the green marker.

An interesting observation is that Hybrid degrades the smallest as illumination increases. Frame differencing and Farneback flow degrade by +70% and +62% from ambient to active fixture conditions, while Detection-only and Hybrid degrade by +55% and +48%, respectively.

Jitter behavior is harder to interpret than accuracy (Table 2). Three observations stand out in particular.

First, Detection-only reports the lowest jitter in every condition. Second, its jitter slope decreases by approximately 45%. Read at face value, this would say detection becomes more stable as lighting becomes harsher, which is hard to understand with Detection-only’s +55% accuracy degradation over the same range. Our hypothesis is that YOLOv8n returns fewer detections under harsher lighting, so the aim point moves less often and our jitter metric, which sums movement per second, goes down even while mean error goes up.

Third, the classical motion baselines move sharply in the opposite direction. Frame difference jitter rises +112% and Farneback flow jumps to +106% as the scene uses both fixture and external dynamic lighting. This is consistent with the expectation that these methods respond to all moving pixels, including the additional motion energy contributed by dynamic lighting.

Hybrid’s jitter slope is essentially flat (+14% from ambient to fixture and external dynamic lighting). Hybrid’s abso-

lute jitter is higher than Detection-only’s. Recall that jitter sums how far the predicted aim moves between consecutive frames, so anything that makes the aim jump more from frame to frame raises this number. Hybrid picks the peak of a motion heatmap built from thresholded frame differences, whereas Detection-only reports the center of a bounding box whose corners are filtered by ByteTrack’s Kalman state, which tends to move smoothly between frames. The deployed system applies no temporal smoothing to Hybrid’s target before issuing DMX commands, so the jitter figures reported here reflect the actual signal that drives the fixture.

5.4. Ablation: Effect of Person-Masking

The Hybrid vs. Frame differencing comparison functions as an ablation on the masking step. Both methods compute a frame difference heatmap, but Hybrid restricts the heatmap to person bounding boxes before finding the peak, so the only design difference between them is the person mask itself.

In the three lower stress conditions, Hybrid is about 8 – 11% more accurate than Frame differencing. Under the fixture and external dynamic lighting setup, the gap widens to around 23%. The only design difference between the two methods is the person mask applied before peak finding, so the widening attributes directly to the masking step. As the scene contributes additional non-person motion (from external dynamic lighting), the value that the mask provides grows. While we do not measure where each method’s heatmap peak actually lands per frame, it is apparent that the masking step accounts for a large portion of the Hybrid method’s effectiveness.

5.5. Processing Speed

All methods were evaluated on the CPU of an M5 Max MacBook Pro laptop. Processing speed was unaffected by lighting condition, varying by approximately 5% across conditions for every method. The FPS performances are summarized in Figure 4. Frame differencing runs at $\sim$ 1900 FPS and Farneback flow at $\sim$ 280 FPS. Neither is bottlenecked by anything other than computational arithmetic. Hybrid sustains 65 – 69 FPS, which is approximately 1.8 $\times$ Detection-only’s $\sim$ 37 FPS. This is likely because Hy-

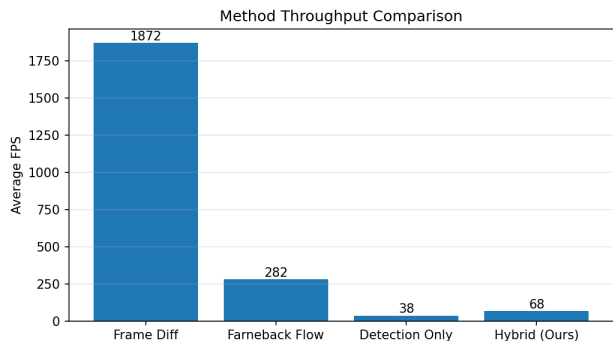


Figure 4. The mean FPS across each condition, measured for each method. Hybrid has almost double the mean FPS of Detection-only. Frame differencing and Farneback flow, being classical methods, experience much higher FPS than the deep learning methods.

brid runs on YOLOv8n every other frame and reuses ByteTrack’s persistent boxes in between, while Detection-only runs the detector every frame.

6. Discussion

The experimental results support a subset of the contribution claims behind VisionBeam. Hybrid is neither the most accurate nor the most stable method on a single performer in stable lighting. However, it is the method that degrades least when the scene becomes chaotic, and it stays fast in real-time while doing so.

6.1. Failure Cases

Just as the literature on optical flow states, the classical baselines (frame differencing and Farneback flow) have no way to tell a moving person from a dynamic light, so once illumination shifts globally or sweeps across a room, their motion signal points at the lighting itself rather than at any performer. The accuracy degradation and jitter increases under dynamic fixture and external lighting confirm this directly. Overall, these methods are not deployable in any environment where the lights are necessary to the venue.

Based on the data, it is harder to agree on a failure mode for the detection-only baseline. It is the best method on absolute accuracy, beating our hybrid approach by around 85 – 115 pixels. The pairing of a -45% jitter slope with a $+55\%$ accuracy slope as illumination becomes harsher is inconsistent if low jitter is taken to mean stability. Therefore, a method whose accuracy is degrading and whose jitter is also dropping is most naturally explained by the predictor returning fewer or staler outputs rather than by it becoming more reliable. We leave the exact reasoning to future work, and conclude for now that detection-only’s effectiveness is measured by its accuracy as opposed to jitter.

The hybrid method produced both promising and confusing results. The bounding boxes acted as a gate removing non-person motion, and frame differencing inside those boxes gave a regional activity measurement without the cost of using action recognition. Removing the detector reduces the system to raw motion, while removing the motion signal reduces it to detection-only. The person-masking mentioned previously is direct evidence that the masking step is what carries the effectiveness, not the underlying motion algorithm.

6.2. Analysis

We do not have a direct measurement of why detection-only outperforms hybrid in absolute accuracy by such a large margin. The data tells us that the gap exists across almost every clip, in almost every lighting condition. It is not apparent which structural features of the two methods produced the gap. We derived two possible explanations consistent with the observations from our experiment.

Bounding Box Center Alignment: Detection returns the geometric center of a tracked bounding box, which on a vertical body box sits near the performer’s vertical midpoint. Hybrid returns the heatmap peak inside that bounding box, which during performance is more likely to land on an arm/leg than on a torso. Under this hypothesis, the gap is essentially the typical distance (in pixels) between the body’s center and its fastest moving body part. Swapping the marker to the performer’s foot might invert or close the gap.

Heatmap Instability: Even if a marker were perfectly placed, the heatmap peak is a frame-by-frame argmax of a noisy intensity field, while the bounding box center is the mean of two coordinates that ByteTrack actively smooths via its Kalman filter. If the heatmap peak fluctuates more between frames than the box center does, hybrid will accumulate frame-by-frame deviations from any stable target. Applying a temporal filter (e.g., an exponential moving average) to hybrid’s output before evaluation could test whether this instability accounts for a meaningful portion of the gap.

6.3. Limitations

Several limitations and caveats should be addressed when reviewing the results holistically.

Ground Truth Marker Placement: In our experiments, the marker was worn on the subject’s chest, while the deployed pipeline aims the beam at the floor beneath the performer (by snapping to the bottom-center of the containing bounding box). The evaluation disables this snap and compares raw predictor outputs in pixel space. Detection-only returns the geometric center of a bounding box, which sits near a chest-worn marker by default. Hybrid returns the motion heatmap peak, which during activity tends to land

Condition	Frame Differencing	Farneback	Detection-only	Hybrid (ours)
Ambient	3369.6 $\pm$ 818.8	2599.5 $\pm$ 1362.9	1103.1 $\pm$ 701.0	2940.2 $\pm$ 1011.2
External (Static)	3224.9 $\pm$ 879.6	2669.0 $\pm$ 1371.4	579.2 $\pm$ 273.5	2719.9 $\pm$ 1047.5
External (Dynamic)	3645.5 $\pm$ 879.3	2570.7 $\pm$ 1338.7	667.6 $\pm$ 373.0	2704.6 $\pm$ 756.2
Fixture + External (Dynamic)	7145.6 $\pm$ 1094.4	5362.7 $\pm$ 931.2	602.8 $\pm$ 239.6	3356.1 $\pm$ 779.7

Table 2. Mean trajectory jitter (in pixels per second) for each method and condition. Detection-only exhibits the lowest jitter over each of the four conditions. Hybrid only performs better than Farneback when both the fixture and dynamic external lighting were activated. Bold marks the best method in each row. We took the mean $\pm$ the sample standard deviation.

on a moving arm/leg rather than the torso. The chest-worn ground truth therefore aligns more naturally with detection-only’s output than with hybrid’s, which connects to one of the discussions in Section 6.2.

Calibration Rigidity: The quadratic pixel-to-angle fit assumes there is a fixed spatial relationship between the camera and the light fixture. Any disruption to either device invalidates the calibration and requires the sampling to be redone. This may be costly in larger settings where equipment is struck and remounted frequently.

Camera Quality: All recordings in this experiment used a simple Logitech webcam with limited range. Under extreme lighting, image quality tends to degrade. A camera with higher dynamic range that is tuned for stage environments may improve performance, particularly under the dynamic external lighting conditions.

Small Dataset: The evaluation comprises twenty 30-second clips with one or two performers. A multi-person dataset with overlapping motion and occlusion would provide a clearer picture of the system’s strengths and weaknesses.

6.4. Ethical and Societal Considerations

In accordance with CVPR Ethics Guidelines, we must consider the broader impacts and potential biases of our system.

VisionBeam relies heavily on the YOLOv8 object detector. It is well-documented that many computer vision models exhibit biased performance across different demographic groups, often performing worse on individuals with darker skin tones. This bias can be worsened in stage environments, where colored lighting and low ambient complicate detection. If the detector systematically fails to identify certain performers, the autonomous lighting will lead to unequal representation on stage. Mitigating this requires ensuring that the underlying detection models are trained on diverse datasets that include varied lighting conditions and skin tones.

The deployment of autonomous tracking systems in public (or semi-public) spaces (e.g. clubs, bars, theaters) raises some privacy concerns. While VisionBeam does not perform facial recognition or store identifying data, the continuous tracking of individuals’ movements constitutes a form

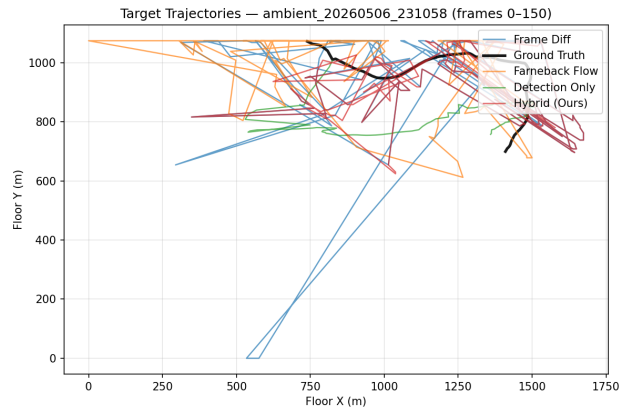


Figure 5. Visualizes the accuracy of each method in tracking the subject, compared to the ground truth marker (represented in black), for the ambient scene. The metrics are taken from the first 150 frames of this specific recording. The classical flow methods are notably worse at smoothly following the ground truth. Hybrid covers a closer distance to the ground truth, but is much more oscillatory than Detection-only.

of surveillance. There is a risk of misuse if such systems are deployed without the knowledge or consent of the participants. Transparent communication about the presence and function of tracking technology is essential to ensure ethical deployment.

7. Conclusion

We presented VisionBeam, an autonomous stage lighting system that uses a hybrid detection and motion pipeline for target selection. Our evaluation across four lighting conditions demonstrates that the hybrid method is not the most accurate in absolute terms, as detection-only wins by a consistent margin. However, it is the most robust. Hy-

brid exhibits the smallest accuracy degradation (+48%) and by far the smallest jitter degradation (+14%) as conditions progress from ambient to dynamic lighting. The classical motion baselines (frame differencing and Farneback flow) more than double in both error and jitter under the same transition, confirming that they are not deployable in dynamic stage environments. Comparing the hybrid approach to strict frame differencing also provides direct evidence that person-masking is the mechanism responsible for this effectiveness. Both methods compute the same heatmap, but masking suppresses the non-person motion that dominates under hostile illumination. Finally, Hybrid maintains real-time performance (~ 67 FPS) under all conditions, comfortably sitting above the 30 FPS deployment floor and approximately $1.8\times$ faster than detection-only. Future work will extend the evaluation to scenarios with three or more people, explore other deep learning methodologies (such as RAFT), and investigate how VisionBeam performs with better hardware.

7.1. Contributions

Ryan contributed to the USB-to-DMX512 pipeline and the motion tracking algorithms. He also was the subject during the testing and recording runs. Tye contributed to the calibration setup and the UI interface used in the calibration stage. Both Ryan and Tye contributed equally to writing up the final paper and creating figures.

7.2. AI Usage Statement

Generative AI tools (Claude Opus 4.7 and Gemini 3.1 Pro) were used for code refactoring suggestions and grammar review. The system design, experiments, analysis, figures, and technical claims were produced and verified by the authors.

References

- [1] G. Farneback. Two-Frame Motion Estimation Based on Polynomial Expansion. In *Proc. 13th Scandinavian Conference on Image Analysis (SCIA)*, LNCS 2749, pp. 363–370, 2003. [2](#)
- [2] D. Fortun, P. Bouthemy, and C. Kervrann. Optical Flow Modeling and Computation: A Survey. *Computer Vision and Image Understanding*, 134:1–21, 2015. [2](#)
- [3] A. Alfarano et al. Estimating Optical Flow: A Comprehensive Review of the State of the Art. 2024. [2](#)
- [4] S. J. Kim, J. M. Frahm, and M. Pollefeys. Robust Motion Estimation Under Varying Illumination. *Image and Vision Computing*, 23(4):365–375, 2005. [2](#)
- [5] Y. Mileva, A. Bruhn, and J. Weickert. Illumination-Robust Variational Optical Flow with Photometric Invariants. In *Proc. DAGM (Pattern Recognition)*, LNCS 4713, pp. 152–162, 2007. [2](#)
- [6] Z. Teed and J. Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. In *Proc. European Conference on Computer Vision (ECCV)*, pp. 402–419, 2020. [2](#)
- [7] S. Agnihotri et al. FlowBench: A Robustness Benchmark for Optical Flow. *OpenReview*, 2024. [2](#)
- [8] G. Jocher, A. Chaurasia, and J. Qiu. Ultralytics YOLOv8. Version 8.0.0, 2023. [2](#)
- [9] Y. Liu et al. Dark-YOLO: A Low-Light Object Detection Algorithm Integrating Multiple Attention Mechanisms. *Applied Sciences*, 15(9):5170, 2025. [2](#)
- [10] H. Liu, Z. Zhang, and B. Feng. POP-YOLOv8: An Object Detection Framework for Partially Occluded Pedestrians in Nighttime Traffic Environments. *Scientific Reports*, 2026. [2](#)
- [11] Y. P. Loh and C. S. Chan. Getting to Know Low-light Images with the Exclusively Dark Dataset. *Computer Vision and Image Understanding*, 178:30–42, 2019. [2](#)
- [12] P. Sun et al. DanceTrack: Multi-Object Tracking in Uniform Appearance and Diverse Motion. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 20961–20970, 2022. [3](#)
- [13] Y. Zhang et al. ByteTrack: Multi-Object Tracking by Associating Every Detection Box. In *Proc. European Conference on Computer Vision (ECCV)*, pp. 1–21, 2022. [3](#)
- [14] J. Gao, H. Zou, F. Zhang, and T.-Y. Wu. An Intelligent Stage Light-based Actor Identification and Positioning System. *International Journal of Information and Computer Security (IJICS)*, 18(1/2):204–218, 2022. [3](#)